

GAYATRI TIRODKAR

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SUMMARY

- I am currently working as **Data Engineer** for past one year before which I have completed my **Post Graduation Diploma in Big Data Analytics** from CDAC in **2015**.
- Passionate about new learning new cutting edge technologies to hone my skills and become a better professional.
- I have worked in challenging and progressive environment by adding value to the organisation by excelling in any role that I take up.

PROFESSIONAL SKILL SET

Ruby	Web Scraping using Ruby, Python	Elastic Search
MongoDB	Nodejs	AWS(EC2, SQS, S3)
Rest API	Postgresql	DynamoDB
Python	Web application development using Sinatra	Image Processing using OCR

EDUCATIONAL DETAILS

CDAC, Pune	Post Graduation Diploma in Big Data Analytics	69.13%
Rajiv Gandhi Institute of Technology, Versova, Mumbai	B.E. in Computer Science	61.23%

WORK EXPERIENCE

CubicTree Technologies Pvt. Ltd.

Data Engineer (2nd of November 2016 - Present)

Role & Responsibilities:

- Web scrapping from discrete websites to extract data using Ruby and Python.
- Store extracted data in specified format in mongodb.
- Migrate data from mongodb according to business requirements into Elastic Search for faster retrieval of the data.
- Query Elastic Search and present the data in PDF, JSON, CSV, XML format for generation of automated reports.
- Bulk insertion of jobs to generate automated reports.

- Generate CSV as per business requirements by querying MongoDB.
- Daily working on the current architecture by detecting and resolving issues if any.
- Handle production issues by writing crawlers and parsers if required in case of changes in the websites or modifying the current scripts if cases or data found missing.
- Data extraction from PDFs, excels and other file formats using ruby as per business requirements.
- Suggest ways and ideas for upgradation and betterment of current architecture.
- Successfully completed multiple and business critical projects giving timely client deliveries and enhancing current system performance as and when required.
- Developed and designed a dashboard giving daily system update, amount of data available and its segregation.
- Currently putting dedicated efforts by fully focusing on the most crucial project and successfully deployed APIs to the client meeting all the possible and feasible requirements. Also learned to modify the requirements to suggest better performance efficient solutions to the client if required.

PROFESSIONAL PROJECTS

1. Title: Information Retrieval Information Extraction (IRIE)

Technologies learned: Ruby, Python, AWS (SQS, S3, EC2), MongoDB

Description:

- The central project of the system which contains multiple crawlers and parsers which are scheduled to execute on daily basis and get the required data in specified format from the designated websites and insert it into MongoDB.
- The process is automated using AWS EC2 for machine creation and AWS SQS is used for automating the job creation by passing in the required parameters to crawl the web pages. The crawled pages are stored in AWS S3 and the parsed data is inserted in MongoDB.
- The parsers written are efficient enough to handle the dynamic nature of the data formats which may vary on daily basis for few websites.
- Once the data is fully loaded into MongoDB the email alerts scripts are executed which forms automated mails to the client giving them updates upto their cases found if any by providing details like when is the case scheduled for next hearing, when, who will be the judge, history of the case for past reference if available etc.
- The automated report about the records inserted, the type and number of cases of each type inserted in the MongoDB is also generated and mailed to the team for daily system updates and makes it feasible to detect and resolve the issues of missing cases if any.

Tasks handled:

- Wrote multiple crawlers and parsers but the major one was ROC data which helped in getting maximum data and resulted in creation of new product solely using the data.
- Fixed crawlers and parsers for few sites when the received data format was changed or the internal working of the websites changed.
- Generated automated emails alert system for the team to get daily updates of the system.

Current Role:

- The major issue currently faced is breaking the CAPTCHAs which are not feasible for the OCR to break.
 - Currently doing the POC to develop an efficient algorithm to break CAPTCHA using Python.
 - Working on writing an universal crawler and parser to scrape the multiple websites and get the desired data.
 - Working on adding case status data in MongoDB.
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2. Title : Heartbeat (System Status Dashboard)

Technologies learned: Ruby, Sinatra, Slim, HTML5, CSS, Bootstrap, Postgresql

Description:

- Dashboard which gives daily updates regarding the total data currently company has, its distribution and visualisation.

Tasks Handled:

- Queried MongoDB to extract only the required data and insert it into Postgresql.
 - Learned to work on Model-View-Controller framework which forms the basic flow for the project.
 - Queried Postgresql to present data on the web application in the form of bars and graphs.
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3. Title: Borrower Profile Reporting System (BPR) :

Technologies learned: Ruby, Nodejs, Elastic Search, Restful API, MongoDB, AWS (S3, EC2), Dynamodb, Postgresql, Redis server

Description :

- This project creates automated report generation which contains a well formatted presentation of the data queried by Elastic Search.
- The report generated is consumed by the clients which helps them to decide whether or not to grant loan to the individual / company for whom the report is generated.
- It gives summarized as well as detailed report in discrete file format like PDF, CSV, JSON, XML which contains company details, cases filed in different courts, defaulter cases details and a lot more.
- The project functions on two major logics: a. Fuzzy and b. City
- Fuzzy logic lists all the matching data cases without applying any filter whereas City logic lists only the cases matched in the specified city.
- The address matcher is also used in order to make report more specific by listing the percentage match on the address if given giving out chances to identify the most possible case.

Tasks completed:

- Migrated only required fields of data from MongoDB to Elastic Search in bulk.
- Wrote address matcher code to match the given address and address found by Elastic Search query to give the match percentage which makes it feasible to identify the most relevant case.
- Worked on automating addition of cities if any city is given which is not found in the city database.
- Added data into report by quering Elastic Search indices.
- Optimised the code by making bulk insertion of the reports.
- Provided Restful API to the clients and modified the code as per their requirements.
- Added summary table giving count of cases of each distinct type found.
- Added ROC company data including different types of defaulters.
- Wrote dynamic crawler and parser to get casestatus details (data not present in MongoDB) from website only when asked.
- Created Internal BPR portal for the team.
- Added remove button and keyword search functionality on the portal.

Current Role:

- Merge casestatus data now added into MongoDB by migrating it to Elastic Search and finally add it into the report.
- Generate summarized report by inserting hyperlinks which opens the case details a when clicked.
- Categorize the individual as the director of some company or the dealer etc. Specifying it as a possibility.
- Generate report in XML format.

4. Title : Uniqueness of data (Mini project)

Technologies learned: MongoDB, Elastic Search, Ruby

Descripton:

- This is an idea for optimization of the search results.
- The data inserted in MongoDB contains multiple similar cases and so the goal is to identify only unique cases which are the recent ones get a list of all the dates of the previous hearings for such cases and their corresponding entry count.
- Migrate the data in Elastic Search which will develop the historic data representation and include it in report.

Task Completed:

- Develop an algorithm for reaching the above goal.

5. Rule Engine: (Currently in initial stage)

Description:

- The project aims to generate report on every check for the given company.
- This is restricted only to the ROC data of the companies and wishes to generate report that contains details only specific to the company, directors, defaulters the charges against them, their valid identification
- This will be given as Restful API to the client as well.

Task Assigned:

- Design and develop the flow and algorithm to reach feasible milestones.
- Study the requirements and form API
- Study the requirements list out all the feasible requirements and develop a plan to reach the end goal.
- List out additional technologies to be required if any and decide which ones from the current ones to be used.
- Develop a flow using the above information.

BASIC INFORMATION

- **Date of Birth:** 8th of May, 1992
- **Languages Known:** English, Hindi, Marathi.
- **Current CTC:** 3 LPA
- **Expected CTC:** 6 LPA
- **Current Location:** Pune, Maharashtra.
- **Relocation / Travel:** Yes.